YEAR 10 SCIENCE
DATA SHEET (EXAMINATION)

Periodic table

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1 H hydrogen 1.008																	2 He helium 4.003
3 Li lithium 6.968	4 Be beryllium 9.012											5 B boron 10.82	6 C carbon 12.01	7 N nitrogen 14.01	8 O oxygen 16.00	9 F fluorine 19.00	10 Ne neon 20.18
11 Na sodium 22.99	12 Mg magnesium 24.31											13 Aℓ aluminium 26.98	14 Si silicon 28.09	15 P phosphorus 30.97	16 S sulfur 32.07	17 Cℓ chlorine 35.45	18 Ar argon 39.95
19 K potassium 39.10	20 Ca calcium 40.08	21 Sc scandium 44.96	22 Ti titanium 47.88	23 V vanadium 50.94	24 Cr chromium 52.00	25 Mn manganese 54.94	26 Fe iron 55.85	27 Co cobalt 58.93	28 Ni nickel 58.69	29 Cu copper 63.55	30 Zn zinc 65.38	31 Ga gallium 69.72	32 Ge germanium 72.59	33 As arsenic 74.92	34 Se selenium 78.96	35 Br bromine 79.90	36 Kr krypton 83.80
37 Rb rubidium 85.47	38 Sr strontium 87.62	39 Y yttrium 88.91	40 Zr zirconium 91.22	41 Nb niobium 92.91	42 Mo molybdenum 95.94	43 Tc technetium	44 Ru ruthenium 101.1	45 Rh rhodium 102.9	46 Pd palladium 106.4	47 Ag silver 107.9	48 Cd cadmium 112.4	49 In indium 114.8	50 Sn tin 118.7	51 Sb antimony 121.8	52 Te tellurium 127.6	53 I iodine 126.9	54 Xe xenon 131.3
55 Cs caesium 132.9	56 Ba barium 137.3	57–71 *La lanthanum 138.9	72 Hf hafnium 178.5	73 Ta tantalum 180.9	74 W tungsten 183.9	75 Re rhenium 186.2	76 Os osmium 190.2	77 Ir iridium 192.2	78 Pt platinum 195.1	79 Au gold 197.0	80 Hg mercury 200.6	81 Tℓ thallium 204.4	82 Pb lead 207.2	83 Bi bismuth 209.0	84 Po polonium	85 At astatine	86 Rn radon
87 Fr francium	88 Ra radium 226.0	89–103 **Ac actinium	104 Rf rutherfordium	105 Db dubnium	106 Sg seaborgium	107 Bh bohrium	108 Hs hassium	109 Mt meitnerium	110 Ds darmstadtium	111 Rg roentgenium	112 Cn copernicium	113 Uut ununtrium	114 Fℓ flerovium	115 Uup ununpentium	116 Lv livermorium	117 Uus ununseptium	118 Uuo ununoctium

* Lanthanide series	58 Ce cerium 140.1	59 Pr praseodymium 140.9	60 Nd neodymium 144.2	61 Pm promethium	62 Sm samarium 150.4	63 Eu europium 152.0	64 Gd gadolinium 157.3	65 Tb terbium 158.9	66 Dy dysprosium 162.5	67 Ho holmium 164.9	68 Er erbium 167.3	69 Tm thulium 168.9	70 Yb ytterbium 173.0	71 Lu lutetium 175.0
** Actinide series	90 Th thorium 232.0	91 Pa protactinium	92 U uranium 238.0	93 Np neptunium	94 Pu plutonium	95 Am americium	96 Cm curium	97 Bk berkelium	98 Cf californium	99 Es einsteinium	100 Fm fermium	101 Md mendelevium	102 No nobelium	103 Lr lawrencium

Key:

Atomic number
Symbol
Name
Standard atomic weight

Solubility rules for ionic solids in water

Soluble in water

Soluble	Exceptions	
	Insoluble	Slightly soluble
Most chlorides	AgCℓ	PbCℓ$_2$
Most bromides	AgBr	PbBr$_2$
Most iodides	AgI, PbI$_2$	
All nitrates	No exceptions	
All ethanoates		
Most sulfates	$SrSO_4$, $BaSO_4$, $PbSO_4$	$CaSO_4$, Ag_2SO_4

Insoluble in water

Insoluble	Exceptions	
	Soluble	Slightly soluble
Most hydroxides	NaOH, KOH, $Ba(OH)_2$ NH_4OH*, AgOH**	$Ca(OH)_2$, $Sr(OH)_2$
Most carbonates	Na_2CO_3, K_2CO_3, $(NH_4)_2CO_3$	
Most phosphates	Na_3PO_4, K_3PO_4, $(NH_4)_3PO_4$	
Most sulfides	Na_2S, K_2S, $(NH_4)_2S$	

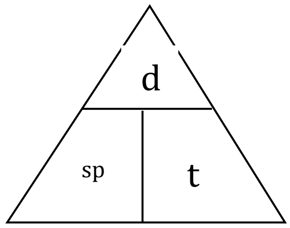
NAMES AND FORMULAE FOR SOME COMMON IONS AND POLYATOMIC IONS

+1 charge	*−1 charge*
lithium, Li^+ sodium, Na^+ potassium, K^+ copper (I) or cuprous, Cu^+ silver, Ag^+	fluoride, F^- chloride, Cl^- bromide, Br^- iodide, I^- hydride, H^-
ammonium, NH_4^+ hydrogen, $H^+_{(aq)}$	hydroxide, OH^- nitrite, NO_2^- nitrate, NO_3^- acetate, CH_3COO^- hydrogencarbonate or bicarbonate, HCO_3^- hydrogensulfate or bisulfate, HSO_4^- chlorate, ClO_3^-
+2 charge	*−2 charge*
manganese Mn^{2+} magnesium, Mg^{2+} calcium, Ca^{2+} barium, Ba^{2+} zinc, Zn^{2+} copper (II) or cupric, Cu^{2+} mercury (II) or mercuric, Hg^{2+} iron (II) or ferrous, Fe^{2+} tin (II) or stannous, Sn^{2+} lead (II) or plumbous, Pb^{2+} nickel(II), Ni^{2+}	oxide, O^{2-} sulfide, S^{2-}
	carbonate, CO_3^{2-} sulfate, SO_4^{2-} sulfite, SO_3^{2-}

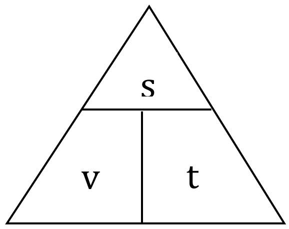
+3 charge	*−3 charge*
aluminium, Al^{3+} iron (III) or ferric, Fe^{3+} chromium, Cr^{3+}	nitride, N^{3-}
	phosphate, PO_4^{3-}
+4 charge	
tin (IV) or stannic, Sn^{4+} lead (IV) or plumbic, Pb^{4+}	

THESE NAMES AND FORMULAE MUST BE MEMORISED

Speed $\text{speed} = \frac{d}{t}$



Velocity $v = \frac{s}{t}$



Acceleration $a = \frac{v - u}{t}$

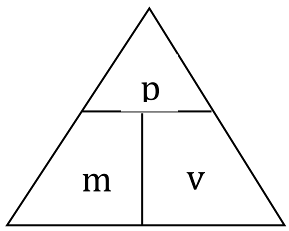
Equations of motion

$$v = u + at$$

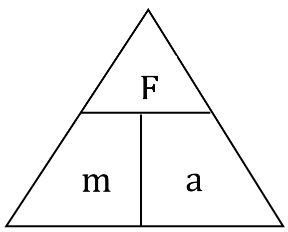
$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Momentum $p = mv$

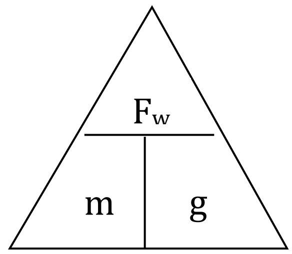


Force $F = ma$



$$F = \frac{mv - mu}{t}$$

Weight $F_w = mg$



Acceleration due to gravity $g = 9.80\ ms^{-2}$

Converting kmh^{-1} to ms^{-1} Divide the speed in kmh^{-1} by 3.6 to get the speed in ms^{-1}.